

A Novel Delta Operator Kalman Filter Design and Convergence Analysis

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Abstract—This paper focuses on the development of a delta operator Kalman filter and its convergence analysis. The delta operator Kalman filter is designed to estimate the state vectors of a delta operator system. Note that the designed delta operator Kalman filter can express both continuous-time and discrete-time cases. Then, the convergence analysis of the delta operator Kalman filter is also investigated by using Lyapunov approach in delta domain. Furthermore, this paper gives fundamental results for the analysis and application of the delta operator Kalman filter as a state observer in an inverted pendulum model. Some experimental results of an inverted pendulum on a laboratory-scale setup are presented to illustrate the effectiveness of the designed Kalman filter and its implementation.

Index Terms—Convergence analysis, delta operator systems, inverted pendulum model, Kalman filter, Lyapunov approach.

I. INTRODUCTION

THE KALMAN filter has been a very powerful tool in control system design since 1960 when Kalman offered his original paper [1]. A robust Kalman filter for systems under norm bounded uncertainties was given in [2]. The Kalman filter was also used for networked data fusion [3]. Kalman filter-based identification of unknown exogenous input of stochastic linear systems was investigated via pseudomeasurement approach in [4]. An optimization method of linear observations for Kalman filter was given based on a generalized water filling theorem in [5]. The Kalman filter was also used in GPS/SINS integrated navigation fusion algorithm in [6]. A two-stage Kalman filter method has been given for chaos systems in [7]. Both a Kalman-filter-based iterative algorithm and a recursive frequency estimator using linear prediction

were considered in [8]. Kalman filter-based fusion algorithm has been applied to estimation problems in [9]. Some results on exact Kalman filtering of polynomial systems have been given in [16]. Robust Kalman filtering for continuous systems with Markovian jumping parameters was given in [18], [19]. The convergence analysis of an extended Kalman filter for nonlinear deterministic systems has been presented in [20]. A problem of optimal linear Kalman filtering with packet losses has been considered in [21].

The Kalman filter has also been widely used in many engineering and scientific areas. Using an extended Kalman filter, the load disturbance rejection response was remarkably improved by compensation of estimated load torque [10]. The Kalman filter for a parallel-structured single-link flexible arm mounted on moving base has been given in [11]. To improve the efficiency of greenhouse climate control, an extended Kalman filter was used in [12]. Kalman filter in time-lapse microscopy, marker-controlled watershed, and tracking using mean-shift have been used for nuclei segmentation in [13]. Detecting unstable fixed points using Kalman filters with constraints has been investigated in [14]. A robust Kalman filter-based subspace tracking algorithm in an impulsive noise environment has been given in [15]. It evaluated the applicability of a Kalman filter to the identification of an autonomous underwater vehicle [17]. The control of an inverted pendulum has been studied and demonstrated in control laboratories in connection with the control of launching the rocket for decades. There is a long list of references related to inverted pendulum control for examples [23]–[25]. By using an extended Kalman filter to estimate the fault-related parameters, a sensor failure detection and identification scheme for a closed-loop inverted pendulum system was described in [22]. In this paper, we use the inverted pendulum model in an experiment to illustrate the effectiveness of the designed Kalman filter and its implementation.

It is well known that discrete systems are obtained by sampling continuous systems and the shorter the sampling period is the better the system performances are. Using traditional shift operator, most of the poles are located in the stable boundary when sampling is fast. Due to this fact, Goodwin constructed delta operator instead of traditional shift operator for sampling continuous systems with short sampling periods [26]. The delta operator model can unify some previous related continuous and discrete systems into the delta operator system framework [27]. By using delta operator, the digital controller design problem for closed-loop stability subject to finite word length implementation was addressed in [28]. A refined digital redesign was proposed to improve the state matching performance by using delta operator [29]. Recently, a robustness stabilization problem for

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delta operator systems with time-varying delays was promoted in [30]. Furthermore, the robust sliding mode control for a class of time-delay delta operator systems was considered in [31]. A delta operator Riccati equation was gained for discrete-time Kalman filter in [32]; however, a detailed delta operator Kalman filter algorithm was not given in the results. A generalization structure of the transposed direct-form II filter structures in shift operator and delta operator was derived in [33]. A least square error criterion based procedure to design 2-D digital filters in delta-domain was presented in [34]. However, to the best of our knowledge, there are few papers considering the convergence of delta operator Kalman filters up to now. Therefore, it is an interesting idea and stimulating to apply a discrete-time Kalman filter in delta domain to estimate the full-state of practical inverted pendulum model.

In this paper, a general delta operator Kalman filter as a state observer is introduced to estimate the full-state vectors of delta operator systems. The designed delta operator Kalman filter can express both continuous-time and discrete-time cases. By using Lyapunov approach, the convergence analysis of the designed delta operator Kalman filter is also given. Some experimental results on a laboratory-scale setup are presented to illustrate the approach and its implementation. This paper is organized as follows. In Section II, the modeling is described by using a delta operator system. Section III gives the main results for the delta operator Kalman filter and its convergence analysis. In Section IV, we give some experimental results. Conclusions are given in Section V.

Notation: In the sequel, if not explicitly stated, matrices are assumed to have compatible dimensions. $\mathbf{R}^n$ denotes the n -dimensional Euclidean space; the notation $X > Y$ ($X \geq Y$) means that the matrix $X - Y$ is positive definite ($X - Y$ is semi-positive definite, respectively); I is the identity matrix of appropriate dimension; For any matrix A , A^T denotes the transpose of matrix A , A^{-1} denotes the inverse of matrix A ; the shorthand $\text{diag}\{M_1 M_2 \dots M_r\}$ denotes a block diagonal matrix with diagonal blocks being the matrices $M_1, M_2, \dots, M_r$; $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ denote the minimum and maximum eigenvalue of the corresponding matrix, respectively. $\|\cdot\|$ denotes the Euclidean norm for vectors or the spectral norms of matrices. $\mathbf{E}(\cdot)$ stands for the mathematical expectation. The symmetric terms in a symmetric matrix are denoted by $*$.

II. PROBLEM FORMULATION

The following continuous system is considered as

$$\dot{x}(t) = A_s x(t) + B_s u(t) + \omega(t) \quad (1)$$

$$y(t) = C_s x(t) + v(t) \quad (2)$$

where $x(t) \in \mathbf{R}^n$ is the plant state, $u(t) \in \mathbf{R}^p$ is the control input, $y(t) \in \mathbf{R}^m$ is the measurement output, $A_s \in \mathbf{R}^{n \times n}$ is the system matrix, $B_s \in \mathbf{R}^{n \times p}$ is the input matrix, $C_s \in \mathbf{R}^{m \times n}$ is the measurement matrix, $\omega(t) \in \mathbf{R}^n$ is the input noise, $v(t) \in \mathbf{R}^m$ is the measurement noise. The two noises are given as $\omega(t_k) \sim (0, Q_s)$ and $v(t_k) \sim (0, R_s)$, where Q_s and R_s are ordinary covariance matrices.

In this paper, we consider a fast reaction case for system (1) and (2). Hence, a small sampling period should be used to meet

the fast reaction case such as in an inverted-pendulum model. By using delta operator approach, the following discrete system is obtained:

$$\delta x(t_k) = A x(t_k) + B u(t_k) + D \omega(t_k) \quad (3)$$

$$y(t_k) = C x(t_k) + v(t_k) \quad (4)$$

in which the delta operator is defined by

$$\delta x(t_k) = \begin{cases} \frac{dx(t)}{dt}, & T_s = 0 \\ \frac{x(t_k + T_s) - x(t_k)}{T_s}, & T_s \neq 0. \end{cases}$$

Remark 1: Form the definition of delta operator, it can be obtained that $\delta x(t_k) \rightarrow \dot{x}(t_k)$ as $T_s \rightarrow 0$ and $\delta x(t_k) = \dot{x}(t_k)$ when $T_s = 0$. When $T_s = 1$, there exists $\delta x(t_k) = x(t_{k+1}) - x(t_k)$ which is equivalent to traditional shift operator. Also, the convergence process of $\delta x(t_k) \rightarrow \dot{x}(t_k)$ is a monotonous process. That is why the sampling period should be less than 1 to obtain the advantage of the delta operator approach.

Moreover, A , B , C and D are constant matrices given as below:

$$A = \frac{e^{A_s T_s} - I}{T_s}, \quad B = \frac{1}{T_s} \int_0^{T_s} e^{A_s(T_s - \tau)} B_s d\tau, \\ C = C_s, \quad D = \frac{1}{T_s} \int_0^{T_s} e^{A_s(T_s - \tau)} d\tau.$$

There exists a set of well-known relations: $\lim_{T_s \rightarrow 0} A = A_s$, $\lim_{T_s \rightarrow 0} B = B_s$, and $\lim_{T_s \rightarrow 0} D = I$.

Remark 2: If we utilize the traditional shift operator, system (1) and (2) can be changed to the following discrete system:

$$x(k+1) = A_z x(k) + B_z u(k) + D_z \omega(k), \\ y(k) = C_z x(k) + v(k)$$

where $A_z = e^{A_s T_s}$, $B_z = \int_0^{T_s} e^{A_s(T_s - s)} B_s ds$, $C_z = C_s$ and $D_z = \int_0^{T_s} e^{A_s(T_s - s)} ds$. When $T_s \rightarrow 0$, there exist $\lim_{T_s \rightarrow 0} A_z = I$ and $\lim_{T_s \rightarrow 0} B_z = 0$. It can be seen that the discretization method for linear systems by using delta operator can possess better numerical properties. The numerical simulation has been given in [30].

Furthermore, the uncorrelated white noises $\omega(t_k) \in \mathbf{R}^n$ and $v(t_k) \in \mathbf{R}^m$ are given as follows:

$$\omega(t_k) \sim (0, Q), \quad Q = Q_s \quad (5)$$

$$v(t_k) \sim (0, R), \quad R = R_s \quad (6)$$

where Q and R are ordinary covariance matrices.

Before ending this section, the following lemmas will be used to develop our main results in the sequel.

Lemma 1 [35]: The continuous-time Kalman filter for the continuous system (1) and (2) is summarized as follows.

- i) For two constants $t > 0$, $\ell > 0$ and the Kronecker delta function $\delta(t)$, the uncorrelated white noises $\omega(t) \in \mathbf{R}^n$ and $v(t) \in \mathbf{R}^m$ with zero mean and two ordinary covariance matrices Q_s and R_s satisfy

$$\mathbf{E}[\omega(t)\omega^T(\ell)] = Q_s \delta(t - \ell)$$

$$\mathbf{E}[v(t)v^T(\ell)] = R_s \delta(t - \ell)$$

$$\mathbf{E}[\omega(t)v^T(\ell)] = 0$$

- ii) The initial state $x(0)$ is independent of $\omega(t) \in \mathbf{R}^n$ and $v(t) \in \mathbf{R}^m$

$$\hat{x}(0) = \mathbf{E}[x(0)]$$

$$P(0) = \mathbf{E}[(\hat{x}(0) - x(0))(\hat{x}(0) - x(0))^T].$$

- iii) The continuous-time Kalman filter equations for the continuous system (1) and (2) are given as

$$K_s = P(t)C_s^T R_s^{-1}$$

$$\dot{\hat{x}}(t) = A_s \hat{x}(t) + B_s u(t) + K_s(y(t) - C_s \hat{x}(t))$$

$$\dot{P}(t) = -P(t)C_s^T R_s C_s P(t) + P(t)A_s^T$$

$$+ A_s P(t) + Q_s.$$

Lemma 2 [35]: The discrete-time Kalman filter is summarized as follows.

- i) For the ordinary covariance matrices Q_z and R_z , the dynamic discrete-time system is given by the following equations:

$$x_k = Fx_{k-1} + Gu_{k-1} + \omega_{k-1}$$

$$y_k = Hx_k + v_k$$

$$\mathbf{E}[\omega_k \omega_j^T] = Q_z \delta_{k-j}$$

$$\mathbf{E}[v_k v_j^T] = R_z \delta_{k-j}$$

$$\mathbf{E}[\omega_k v_j^T] = 0.$$

- ii) The Kalman filter is initialized as follows:

$$\hat{x}_0^+ = \mathbf{E}[x_0]$$

$$P_0^+ = \mathbf{E}[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T].$$

- iii) The discrete-time Kalman filter is given by the following equations, which are computed for each time step $k = 1, 2, \dots$:

$$P_k^- = F P_{k-1}^+ F^T + Q_z$$

$$K_k = P_k^- H^T (H P_k^- H^T + R_z)^{-1}$$

$$= P_k^+ H^T R_z^{-1}$$

$$\hat{x}_k^- = F \hat{x}_{k-1}^+ + G u_{k-1}$$

$$= \text{a priori state estimate}$$

$$\hat{x}_k^+ = \hat{x}_k^- + K_k (y_k - H \hat{x}_k^-)$$

$$= \text{a posteriori state estimate}$$

$$P_k^+ = (I - K_k H) P_k^- (I - K_k H)^T + K_k R_z K_k^T$$

$$= [(P_k^-)^{-1} + H^T R_z^{-1} H]^{-1}$$

$$= (I - K_k H) P_k^-.$$

III. MAIN RESULTS

A. Kalman Filter Design

In this section, a discrete-time Kalman filter called delta operator Kalman filter for system (3) and (4) is introduced.

Theorem 1: A single algorithm of Kalman filter named delta operator Kalman filter is given for the delta operator system

- (3) and (4) with white noise processes (5) and (6) and initial condition

$$\hat{x}(0) = \mathbf{E}[x(0)]$$

$$P(0) = \mathbf{E}[(\hat{x}(0) - x(0))(\hat{x}(0) - x(0))^T]$$

which operates recursively performing a single cycle each time and a new set of measurements becomes available. The delta operator Kalman filter is given by the following equations:

$$K(t_k) = P(t_k)C^T(CP(t_k)C^T T_s + R)^{-1} \quad (7)$$

$$\delta \hat{x}(t_k) = A \hat{x}(t_k) + B u(t_k) + K(t_k)[y(t_k) - C \hat{x}(t_k)]$$

$$- T_s K(t_k) C [A \hat{x}(t_k) + B u(t_k)] \quad (8)$$

$$\delta P(t_k) = -K(t_k)C P(t_k) + A P(t_k) + P(t_k)A^T$$

$$+ Q + T_s A(I - K(t_k)C T_s) P(t_k)A^T$$

$$- T_s [A K(t_k)C P(t_k) + K(t_k)C P(t_k)A^T]. \quad (9)$$

Each iteration propagates the estimate from the time of last measurement is obtained to the current time.

Proof: Transforming the delta operator system (3) and (4), we have the following discrete system:

$$x_k = F x_{k-1} + G u_{k-1} + \Lambda \omega_{k-1} \quad (10)$$

$$y_k = H x_k + v_k \quad (11)$$

where $x_k = x(t_k)$, $\omega_{k-1} = \omega(t_{k-1})$, $v_k = v(t_k)$, $F = (I + A T_s)$, $G = B T_s$, $\Lambda = D T_s$ and $H = C$. The uncorrelated white noises $\omega_k \in \mathbf{R}^n$ and $v_k \in \mathbf{R}^m$ are given as follows:

$$\omega_k \sim (0, Q_z), \quad Q_z = T_s Q$$

$$v_k \sim (0, R_z), \quad R_z = \frac{R}{T_s}.$$

By Lemma 2, the following discrete-time Kalman filter gain for system (10) and (11) is given as

$$K_k = P_k^- H^T (H P_k^- H^T + R_z)^{-1}.$$

Letting $K(t_k) = K_k / T_s$, it can be derived that

$$K_k = P_k^- C^T \left(C P_k^- C^T + \frac{R}{T_s} \right)^{-1}$$

$$K(t_k) = P_k^- C^T (C P_k^- C^T T_s + R)^{-1}.$$

Using Lemma 2 again, the estimation-error covariances are

$$P_k^+ = (I - K_k H) P_k^-$$

$$P_{k+1}^- = F P_k^+ F^T + Q_z.$$

Note that P_{k+1}^- can be rewritten as

$$P_{k+1}^- = (I + A T_s) P_k^+ (I + A T_s)^T + T_s Q$$

$$= P_k^+ + T_s (A P_k^+ + P_k^+ A^T + Q)$$

$$+ T_s^2 A P_k^+ A^T.$$

Substituting P_k^- gives

$$P_{k+1}^- = (I - K_k C) P_k^- + T_s A (I - K_k C) P_k^-$$

$$+ T_s [(I - K_k C) P_k^- A^T + Q]$$

$$+ T_s^2 A (I - K_k H) P_k^- A^T.$$

Subtracting P_k^- from both sides and divided by T_s , we have

$$\begin{aligned} \frac{P_{k+1}^- - P_k^-}{T_s} = & -\frac{K_k C P_k^-}{T_s} + A P_k^- - A K_k C P_k^- \\ & + P_k^- A^T - K_k C P_k^- A^T + Q \\ & + T_s A(I - K_k H) P_k^- A^T. \end{aligned}$$

Letting $P(t_k) = P_k^-$ and $\delta P(t_k) = (P_{k+1}^- - P_k^-)/T_s$, there exists

$$\delta P(t_k) = -K(t_k) C P(t_k) + A P(t_k) + P(t_k) A^T + Q + \gamma(T_s) \quad (12)$$

with

$$\gamma(T_s) = T_s [A(I - K(t_k) C T_s) P(t_k) A^T - A K(t_k) C P(t_k) - K(t_k) C P(t_k) A^T].$$

From Lemma 2, the Kalman filter for $\hat{x}(t_k)$ is

$$\begin{aligned} \hat{x}_k^- &= F \hat{x}_{k-1}^+ + G u_{k-1} \\ \hat{x}_k^+ &= \hat{x}_k^- + K_k (y_k - H \hat{x}_k^-). \end{aligned}$$

The above measurement update equation can be rewritten as

$$\begin{aligned} \hat{x}_k^+ &= F \hat{x}_{k-1}^+ + G u_{k-1} \\ &+ K_k (y_k - H F \hat{x}_{k-1}^+ - H G u_{k-1}) \\ &= K_k (y_k - C \hat{x}_{k-1}^+ - T_s C A \hat{x}_{k-1}^+ - T_s C B u_{k-1}) \\ &+ (I + A T_s) \hat{x}_{k-1}^+ + B T_s u_{k-1}. \end{aligned}$$

Subtracting $\hat{x}_{k-1}^+$ from both sides, divided by T_s , we obtain

$$\begin{aligned} \frac{\hat{x}_k^+ - \hat{x}_{k-1}^+}{T_s} &= A \hat{x}_{k-1}^+ + B u_{k-1} \\ &+ K(t_k) (y_k - C \hat{x}_{k-1}^+ - T_s C A \hat{x}_{k-1}^+ - T_s C B u_{k-1}). \end{aligned}$$

Hence, there exists

$$\begin{aligned} \delta \hat{x}(t_k) &= A \hat{x}(t_k) + B u(t_k) \\ &+ K(t_k) [y(t_k) - C \hat{x}(t_k)] \\ &- T_s K(t_k) C [A \hat{x}(t_k) + B u(t_k)]. \end{aligned}$$

Lastly, we have gotten the (7)–(9) which can be used to integrate the state estimate in delta operator Kalman filter.

This completes the proof. $\blacksquare$

Remark 3: Note that (12) for $P(t_k)$ is called a delta operator Riccati equation, which can be used to compute the estimation-error covariance for the delta operator Kalman filter (7)–(9). Similar delta operator Riccati equation result has also been derived in [32]. However, the results in [32] did not provide detailed discrete-time Kalman filter for delta operator systems.

Remark 4: The delta operator Kalman filter in Theorem 1 can express both continuous-time and discrete-time cases. On one hand, we note from the characteristic of delta operator systems and the following equations:

$$\begin{aligned} K_s &= \lim_{T_s \rightarrow 0} P(t_k) C^T (C P(t_k) C^T T_s + R)^{-1} \\ &= P(t_k) C_s^T R_s^{-1} \end{aligned}$$

$$\begin{aligned} \dot{\hat{x}}(t) &= \lim_{T_s \rightarrow 0} \delta \hat{x}(t_k) \\ &= A_s \hat{x}(t) + B_s u(t) + K_s (y(t) - C_s \hat{x}(t)) \\ \dot{P}(t) &= \lim_{T_s \rightarrow 0} \delta P(t_k) \\ &= -P(t) C_s^T R_s C_s P(t) \\ &\quad + A_s P(t) + P(t) A_s^T + Q_s \end{aligned}$$

that when the sampling period T_s tends to 0 in system (3) and (4), the delta operator Kalman filter (7)–(9) in Theorem 1 reduces to the standard continuous-time Kalman filter in Lemma 1 for continuous system (1) and (2). On the other hand, a traditional discrete-time Kalman filter is obtained when sampling period $T_s = 1$ from the proof of Theorem 1.

Remark 5: The Kalman filter in Lemma 2 is designed for linear invariant discrete systems. Hence, the Kalman filter in Lemma 2 can not be applied for delta operator systems directly. Moreover, the sampling-period in the delta operator Kalman filter (7)–(9) is an explicit parameter. If we replace the invariant sampling period T_s by a time-varying sampling period in (7)–(9), the delta operator Kalman filter is also effective and the Kalman filter in Lemma 2 is hardly used in this case.

B. Convergence Analysis

In this section, we give an approach for setting up the convergence analysis of the delta operator Kalman filter (7)–(9) for system (3) and (4) which is equivalent to system (10) and (11). By considering the state feedback control law $u_k = K x_k$, system (10) and (11) is changed to the following closed-loop system:

$$x_{k+1} = F_K x_k + \varpi_k \quad (13)$$

$$y_k = H x_k + v_k \quad (14)$$

where $\varpi_k = \Lambda \omega_k$ and $F_K = F + G K$, which implies that F_K is a bounded nonsingular matrix. According to system (13) and (14), the delta operator Kalman filter (7)–(9) can be written as the following form:

1) measurement update

$$K_{k+1} = P_{k+1}^- H^T (H P_{k+1}^- H^T + R)^{-1} \quad (15)$$

$$\hat{x}_{k+1}^+ = \hat{x}_{k+1}^- + K_{k+1} (y_{k+1} - H \hat{x}_{k+1}^-) \quad (16)$$

$$P_{k+1}^+ = (I - K_{k+1} H) P_{k+1}^-. \quad (17)$$

2) time update

$$\hat{x}_{k+1}^- = F_K \hat{x}_k^+ \quad (18)$$

$$P_{k+1}^- = F_K P_k^+ F_K^T + Q. \quad (19)$$

It has been shown from the proof of Theorem 1 that (7)–(9) are equivalent to (15)–(19) with $x(t_k) = x_k$. Hence, the objective in this part is to show $\hat{x}_k^+ \rightarrow x_k$ as $k \rightarrow \infty$.

Firstly, we will show that a Lyapunov functional set $\{V_k | k=1, 2, \dots\}$ is a decreasing sequence.

Theorem 2: We let

$$\hat{x}_{k+1}^+ = x_{k+1} - \hat{x}_{k+1}^+$$

$$\hat{x}_{k+1}^- = x_{k+1} - \hat{x}_{k+1}^-$$

be the state error estimation and the state prediction error vectors, respectively. The Lyapunov functional set

$$\left\{ V_k = (\tilde{x}_k^+)^T (P_k^+)^{-1} \tilde{x}_k^+ |_{k=1,2,\dots} \right\}$$

is a decreasing sequence, if the following inequalities:

$$-R_{k+1}^{-1} + R^{-1} H P_{k+1}^+ H^T R^{-1} \leq 0 \quad (20)$$

and

$$F_K^T (F_K P_k^+ F_K^T + Q)^{-1} F_K - (P_k^+)^{-1} \leq 0 \quad (21)$$

hold.

Proof: Choose a Lyapunov functional as follows:

$$V_{k+1}^+ = (\tilde{x}_{k+1}^+)^T (P_{k+1}^+)^{-1} \tilde{x}_{k+1}^+. \quad (22)$$

Setting $e_{k+1} = y_{k+1} - H \hat{x}_{k+1}^-$, we have

$$\begin{aligned} e_{k+1} &\approx H x_{k+1} - H \hat{x}_{k+1}^- = H \tilde{x}_{k+1}^- \\ &\text{and} \\ \tilde{x}_{k+1}^- &\approx F_K x_k - F_K \hat{x}_k^- = F_K \tilde{x}_k^-. \end{aligned}$$

By subtracting both sides of (16) from x_{k+1} , we obtain

$$\tilde{x}_{k+1}^+ = \tilde{x}_{k+1}^- - P_{k+1}^- H^T (H P_{k+1}^- H^T + R)^{-1} e_{k+1}.$$

Based on (15), there exist

$$\begin{aligned} K_{k+1} (H P_{k+1}^- H^T + R) &= P_{k+1}^- H^T \\ K_{k+1} H P_{k+1}^- H^T + K_{k+1} R &= P_{k+1}^- H^T \\ K_{k+1} H P_{k+1}^- H^T - P_{k+1}^- H^T &= -K_{k+1} R \\ (K_{k+1} H - I) P_{k+1}^- H^T &= -K_{k+1} R. \end{aligned}$$

Considering (17), we have

$$\begin{aligned} -P_{k+1}^+ H^T &= -K_{k+1} R \\ P_{k+1}^+ H^T R^{-1} &= K_{k+1}. \end{aligned}$$

Substituting K_{k+1} into (17), it can be obtained

$$P_{k+1}^+ = (I - P_{k+1}^+ H^T R^{-1} H) P_{k+1}^-.$$

Pre-multiplying $(P_{k+1}^+)^{-1}$ and post-multiplying $(P_{k+1}^-)^{-1}$ to the above equation, we get

$$(P_{k+1}^+)^{-1} = (P_{k+1}^-)^{-1} + H^T R^{-1} H. \quad (23)$$

Following the above equations, we have

$$\begin{aligned} V_{k+1}^+ &= (\tilde{x}_{k+1}^- - P_{k+1}^+ H^T R^{-1} e_{k+1})^T (P_{k+1}^+)^{-1} \\ &\quad \times (\tilde{x}_{k+1}^- - P_{k+1}^+ H^T R^{-1} e_{k+1}) \\ &= (\tilde{x}_{k+1}^-)^T (P_{k+1}^+)^{-1} \tilde{x}_{k+1}^- \\ &\quad - 2(\tilde{x}_{k+1}^-)^T H^T R^{-1} e_{k+1} \\ &\quad + e_{k+1}^T R^{-1} H P_{k+1}^+ H^T R^{-1} e_{k+1} \\ &= (\tilde{x}_{k+1}^-)^T (P_{k+1}^-)^{-1} \tilde{x}_{k+1}^- \end{aligned}$$

$$+ e_{k+1}^T (-R^{-1} + R^{-1} H P_{k+1}^+ H^T R^{-1}) e_{k+1}.$$

In the following, V_{k+1}^- is defined as

$$\begin{aligned} V_{k+1}^- &= (\tilde{x}_{k+1}^-)^T (P_{k+1}^-)^{-1} \tilde{x}_{k+1}^- \\ &\approx (\tilde{x}_k^+)^T F_K^T (P_{k+1}^-)^{-1} F_K \tilde{x}_k^+ \\ &= (\tilde{x}_k^+)^T F_K^T (F_K P_k^+ F_K^T + Q)^{-1} F_K \tilde{x}_k^+. \end{aligned}$$

That is,

$$\begin{aligned} V_{k+1}^+ - V_k^+ &= (V_{k+1}^+ - V_{k+1}^-) + (V_{k+1}^- - V_k^+) \\ &= (V_{k+1}^+ - (\tilde{x}_{k+1}^-)^T (P_{k+1}^-)^{-1} \tilde{x}_{k+1}^-) \\ &\quad + ((\tilde{x}_k^+)^T F_K^T (F_K P_k^+ F_K^T + Q)^{-1} F_K \tilde{x}_k^+ - V_k^+) \\ &= e_{k+1}^T (-R^{-1} + R^{-1} H P_{k+1}^+ H^T R^{-1}) e_{k+1} \\ &\quad + (\tilde{x}_k^+)^T [F_K^T (F_K P_k^+ F_K^T + Q_k)^{-1} F_K - (P_k^+)^{-1}] \tilde{x}_k^+. \end{aligned}$$

A decreasing sequence $\{V_k |_{k=1,2,\dots}\}$ means that

$$V_{k+1}^+ - V_k^+ \leq 0.$$

A sufficient condition to ensure that is

$$\begin{aligned} -R^{-1} + R^{-1} H P_{k+1}^+ H^T R^{-1} &\leq 0 \\ F_K^T (F_K P_k^+ F_K^T + Q_k)^{-1} F_K - (P_k^+)^{-1} &\leq 0. \end{aligned}$$

This proof is finished. $\blacksquare$

Even if $\{V_k |_{k=1,2,\dots}\}$ is a decreasing sequence, the convergence of the delta operator Kalman filter (7)–(9) is not ensured in Theorem 2.

Assume that system (13) and (14) satisfies the N -observability rank condition and is N -observable. We define the following manner:

$$\begin{aligned} Y_k &= \begin{bmatrix} y_{k-N} \\ y_{k-N+1} \\ \vdots \\ y_k \end{bmatrix} \approx \begin{bmatrix} H F_K^{k-N} x_k \\ H F_K^{k-N+1} x_k \\ \vdots \\ H x_k \end{bmatrix} \\ &= \begin{bmatrix} H F_K^{k-N} \\ H F_K^{k-N+1} \\ \vdots \\ H \end{bmatrix} x_k = \mathcal{O}_k(k-N, k) x_k. \end{aligned} \quad (24)$$

Due to F_K is a bounded nonsingular matrix, there exist positive real numbers η_1 and η_2 such that for all $k \geq N$ and finite $N \geq 0$, the following Gramian matrix inequality holds:

$$\eta_1 I_n \leq L_{k-N}(\mathcal{O}, \mathcal{R}) \leq \eta_2 I_n \quad (25)$$

with

$$\begin{aligned} \mathcal{R}_k(k-N, k) &= \text{diag}\{R^{-1}, R^{-1}, \dots, R^{-1}\} \\ L_{k-N}(\mathcal{O}, \mathcal{R}) &= \mathcal{O}_k^T(k-N, k) \mathcal{R}_k(k-N, k) \\ &\quad \times \mathcal{O}_k(k-N, k). \end{aligned}$$

Then, the delta operator system (3) and (4) (and its associated the delta operator Kalman filter (7)–(9) for $\hat{x}_k^-$ and $\hat{x}_k^+$ sufficiently close to the true state x_k) is said to be reconstructible.

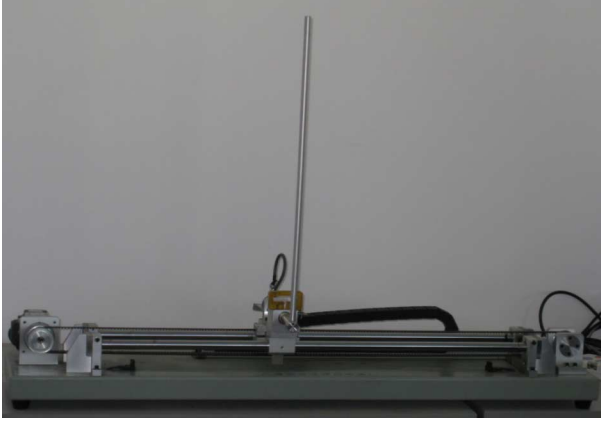


Fig. 1. Inverted pendulum construction.

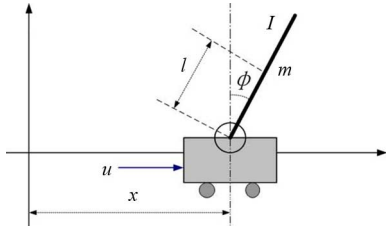


Fig. 2. Inverted pendulum model.

Lemma 3 [20]: If we assume that system (10) and (11) is reconstructible with (25), it is obtained that

$$\lim_{k \rightarrow \infty} \lambda_{\min}\{(P_k^+)^{-1}\} = \infty \quad (26)$$

and

$$\sup_k \lim_{k \rightarrow \infty} \frac{\lambda_{\max}\{(P_k^+)^{-1}\}}{\lambda_{\min}\{(P_k^+)^{-1}\}} < \infty. \quad (27)$$

It is easy to get Lemma 3 from the third lemma in [20] which includes the linear case such as system (10) and (11).

Theorem 3: Suppose that (20), (21) and (25) hold, then the delta operator Kalman filter (7)–(9) used as an observer for the delta operator system (3) and (4), ensures that

$$\lim_{k \rightarrow \infty} (x_k - \hat{x}_k^+) = 0$$

Proof: From (23) and F_K is a bounded nonsingular matrix, it is easy to show that

$$(P_{k+1}^+)^{-1} = L_{k+1}(\mathcal{O}, \mathcal{R}) + \Psi(0, k)$$

with

$$\begin{aligned} \Psi(0, k) &= (F_K^{-k-1})^T P_0^{-1} F_K^{-k-1} \\ L_{k+1}(\mathcal{O}, \mathcal{R}) &= \mathcal{O}_k^T(1, k+1) \mathcal{R}_k(1, k+1) \mathcal{O}_k(1, k+1). \end{aligned}$$

On a horizon time kN , if the constructibility Gramian matrix $\mathcal{O}_k^T(1, kN) \mathcal{R}_k(1, kN) \mathcal{O}_k(1, kN)$ is decomposed into k block matrices of horizon N , we obtain

$$(P_{kN}^+)^{-1} = \sum_{i=1}^k [\mathcal{O}_k^T((i-1)N+1, iN)$$

 TABLE I
PARAMETERS USED IN THE INVERTED PENDULUM MODEL

Symbol	Meaning
m	Mass of the pendulum rod
l	Length from the pendulum's axis to its centroid
I	Inertia of the pendulum rod
g	Gravitational constant
ϕ	Angle between the pendulum rod and the vertical
x	Horizontal displacement of the cart
u	Control input

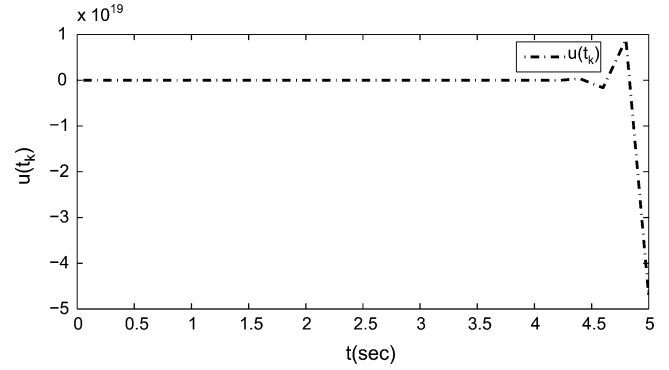


Fig. 3. Input of the inverted-pendulum model for sampling period is 0.2 s.

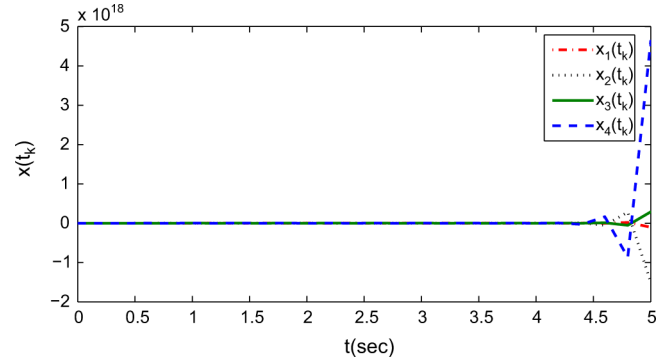


Fig. 4. States of the inverted-pendulum model for sampling period is 0.2 s.

$$\begin{aligned} &\times \mathcal{R}_k((i-1)N+1, iN) \mathcal{O}_k((i-1)N+1, iN)] \\ &+ \Psi(0, kN-1). \end{aligned}$$

Since $(P_{kN}^+)^{-1} - \Psi(0, kN-1)$ is the sum of k constructibility Gramian matrix and under (25), it can be obtained that

$$0 < k\eta_1 \leq \lambda \left\{ (P_{kN}^+)^{-1} - \Psi(0, kN-1) \right\} \leq k\eta_2 \quad (28)$$

where λ represents the eigenvalue symbol of $(P_{kN}^+)^{-1} - \Psi(0, kN-1)$. It is easy to see from (28) that $\lambda \left\{ (P_{kN}^+)^{-1} - \Psi(0, kN-1) \right\}$ is bounded by $k\eta_1$ and $k\eta_2$. Therefore, we have that $k\eta_1 \rightarrow \infty$ and $k\eta_2 \rightarrow \infty$ as $k \rightarrow \infty$.

On the other hand, there exists

$$\frac{V_k^+}{\text{tr}\{(P_k^+)^{-1}\}} > \frac{\lambda_{\min}\{(P_k^+)^{-1}\} (\tilde{x}_k^+)^T \tilde{x}_k^+}{n \lambda_{\max}\{(P_k^+)^{-1}\}} > 0.$$

According to (26) in Lemma 3, we obtain

$$\lim_{k \rightarrow \infty} \text{tr}\{(P_k^+)^{-1}\} = \infty.$$

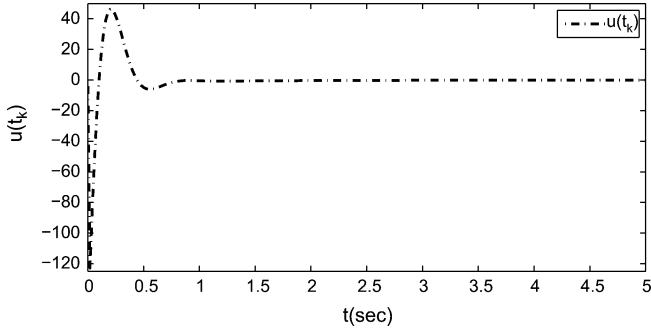


Fig. 5. Input of the inverted-pendulum model for sampling period is 0.02 s.

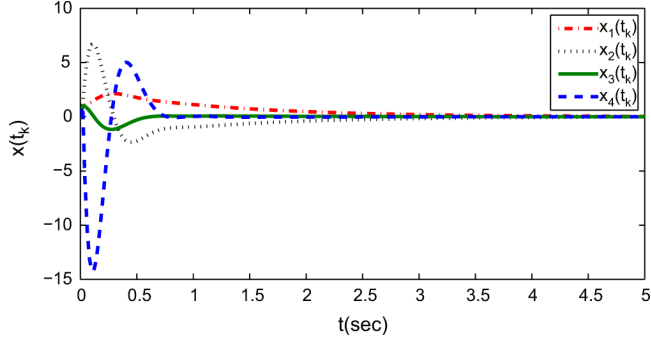


Fig. 6. States of the inverted-pendulum model for sampling period is 0.02 s.

Then, it is obtained that

$$\lim_{k \rightarrow \infty} \frac{\lambda_{\min} \left\{ (P_k^+)^{-1} \right\} (\hat{x}_k^+)^T \hat{x}_k^+}{n \lambda_{\max} \left\{ (P_k^+)^{-1} \right\}} = 0.$$

Thus, (27) yields to

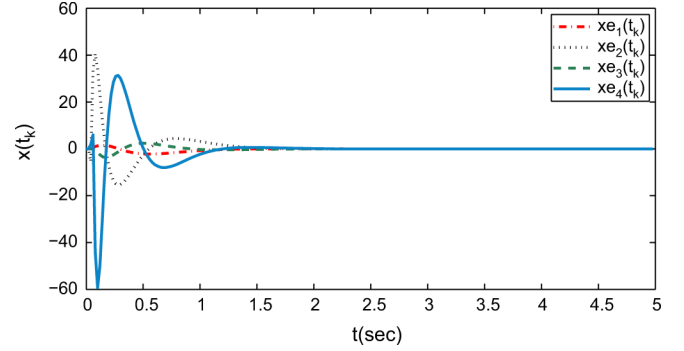
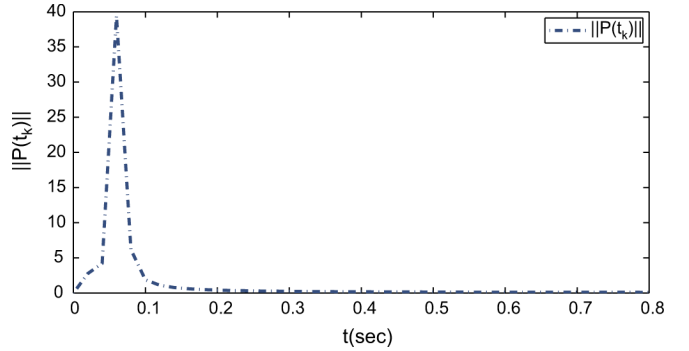
$$\lim_{k \rightarrow \infty} \tilde{x}_k^+ = 0.$$

which means that $\lim_{k \rightarrow \infty} (x_k - \hat{x}_k^+) = 0$.

This completes the proof. ■

Remark 6: It is well known that state feedback control method can not be used in general for the reason of that full states are hardly available such as in inverted pendulum model. Kalman filter as a state observer proves an effective technique to estimate the system states. In some practical systems [36], a continuous-time Kalman filter instead of the discrete-time case was used to accurately estimate system state. However, continuous-time Kalman filter can not be applied in a realistic situation for the application of digital computers. Therefore, we study how to apply delta operator Kalman filter for estimating system state of inverted pendulum model in the following experiment.

Remark 7: The main contribution of this paper is the designed delta operator Kalman filter and its convergence analysis. Some results have been investigated on the problem of Kalman filter in delta domain such as in [37]. The convergence analysis of this kind of delta operator Kalman filters is also an interesting idea to consider. The inverted pendulum control problem is challenging because it may go unstable for low sampling frequencies. In next section, experimental results on inverted pendulum control problem will be given to show the effectiveness of the designed delta operator Kalman filter in this paper.

Fig. 7. Numerical simulation of estimated state vector $\hat{x}(t_k)$.Fig. 8. Numerical simulation of rate of convergence of $\|P(t_k)\|$.

IV. EXPERIMENTAL RESULTS

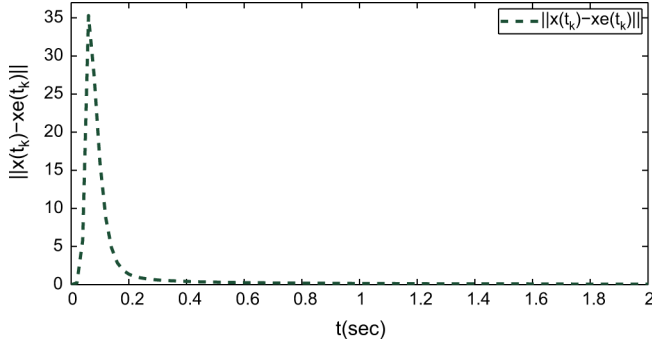
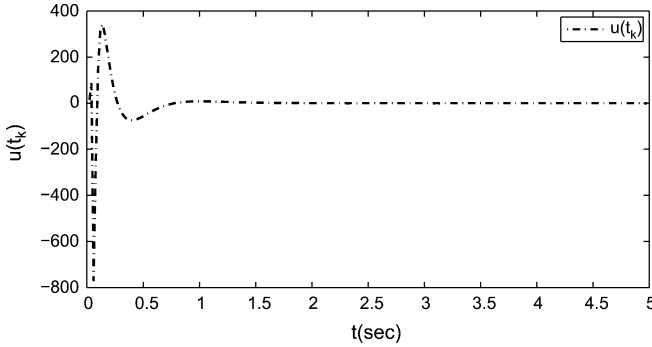
In order to show the effectiveness of the designed delta operator Kalman filter in Theorem 1 and convergence analysis in Theorems 2 and 3, we consider the control of an inverted pendulum model via a practical experiment.

1) *Experiment:* The difference between the real-time simulation and real-time experiments is that the plant model of the inverted pendulum model is replaced by D/A block Dac9, A/D block Adc9 and the real inverted-pendulum system. The real inverted-pendulum model considered here consists of the cart that is moving on a rail and the pole that is mounted on the cart via the pole shaft so that the pendulum can swing freely in the vertical plane. The inverted-pendulum model used in this experiment is shown in Fig. 1.

The translation of the cart is enabled by the dc motor. The shaft of the motor is connected with the cart by a thin steel wire. Control algorithm and data acquisition and visualization are realized with MathWorks Matlab, Simulink, and Real-Time xPC target products on a personal computer with the 12-bit AD/DA converter. The two blocks Dac9 and Adc9 are the driver of the A/D and D/A channels in the embedded system and are designed in Matlab S-Function. Because of the limited pendulum-rail length (0.5 m), the pendulum-cart movement is also limited (0.25 m).

The physical structure of the inverted pendulum on a cart is shown in Fig. 2.

In addition, the surface is friction free for model building simplicity. Symbols meaning in Fig. 2 are shown in Table I, where m , l , I , and g are constants, ϕ , $\dot{x}$, and $\dot{\phi}$ are variables. We let state variables be x , $\dot{x}$, ϕ , and $\dot{\phi}$. They correspond to the horizontal position, horizontal velocity of the cart, angle, and angle velocity of the pole. Because the translation of the cart is enabled


 Fig. 9. Numerical simulation of rate of convergence of $\|x(t_k) - \hat{x}(t_k)\|$.

 Fig. 10. Numerical simulation of control input $u(t_k)$.

by a dc motor, acceleration variable has to be used as a control input variable, which means $\mathbf{u} = -\ddot{\mathbf{x}}$, where “ $-$ ” is based on the turning direction of dc motor.

The equation of motion can be obtained as follows:

$$(I + ml^2) \ddot{\phi} - mgl\phi = ml\ddot{x}.$$

For the swing link with uniform distribution quality, we have the following equation of inertia as

$$I = \frac{1}{3}ml^2.$$

Based on above two equations, we have

$$\left(\frac{1}{3}ml^2 + ml^2\right) \ddot{\phi} - mgl\phi = mlmb\ddot{x}$$

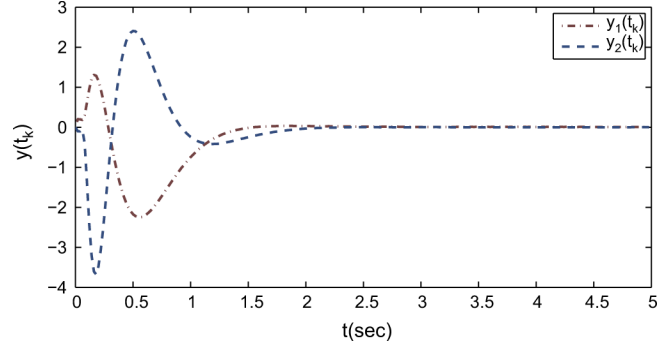
which is changed to

$$\ddot{\phi} = \frac{3g}{4l}\phi + \frac{3}{4l}\ddot{x}.$$

Let the state variables be $\mathbf{x} = x_1(t)$, $\dot{\mathbf{x}} = x_2(t)$, $\phi = x_3(t)$, and $\dot{\phi} = x_4(t)$ and input variable be $u(t) = \mathbf{u}$. Letting $g = 9.81 \text{ m/s}^2$ and $l = 0.25 \text{ m}$, the following continuous system is obtained to describe the inverted pendulum model:

$$\dot{\mathbf{x}}(t) = A_s \mathbf{x}(t) + B_s u(t) + \omega(t) \quad (29)$$

$$y(t) = C_s \mathbf{x}(t) + v(t) \quad (30)$$


 Fig. 11. Numerical simulation of output of $y(t_k)$.

where

$$A_s = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 29.4 & 0 \end{bmatrix}, \quad B_s = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 3 \end{bmatrix},$$

$$C_s = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

It can be seen from $|sI - A_s| = 0$ that there exist $s_1 = s_2 = 0$, $s_3 = -5.4222$ and $s_4 = 5.4222$. The linear system (29) and (30) of the inverted pendulum model is unstable. Furthermore, the controllability matrix is

$$\mathbf{Q}_s = [B_s \quad A_s B_s \quad A_s^2 B_s \quad A_s^3 B_s] \quad (31)$$

and $\text{rank}(\mathbf{Q}_s) = 4$, so the linear model of the inverted pendulum is controllable.

In the following, the linear-quadratic regulator (LQR) is used for the calculation of the optimal gain matrix K such that the following state-feedback control law:

$$u(t) = Kx(t)$$

minimizes the cost function

$$J_{LQG} = \int (x^T(t)Qx(t) + u^T(t)Ru(t))dt.$$

For choosing the appropriate matrices as

$$Q = \begin{bmatrix} 1000 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 200 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad R = [1]$$

we can adjust the state-feedback LQR controller K as follows:

$$K = [-31.623 \quad -20.151 \quad -72.718 \quad -13.155]. \quad (32)$$

The stability of the inverted-pendulum model is guaranteed for the reason of that the LQR design for the proposed state-feedback controller is used.

The inverted-pendulum model is a continuous system with a rapid response. The following results will show that the inverted-pendulum system is only fit for small sampling periods. Firstly, by using the state-feedback controller (32), choosing the

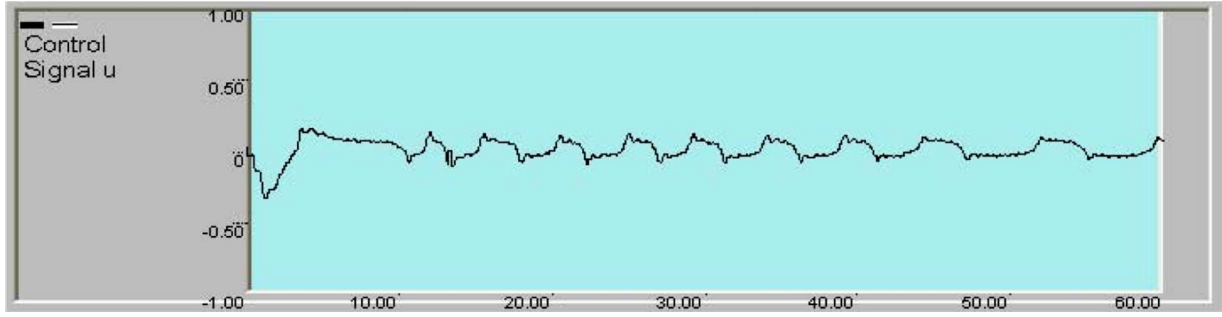


Fig. 12. Experimental result of the control signal.

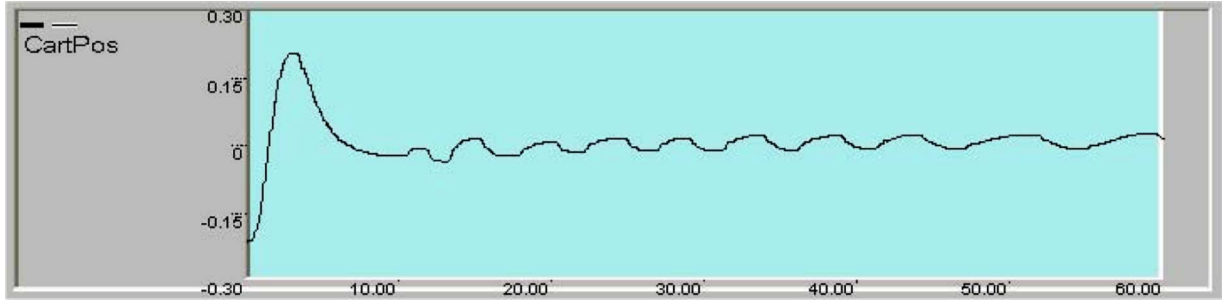


Fig. 13. Experimental result of the cart position.

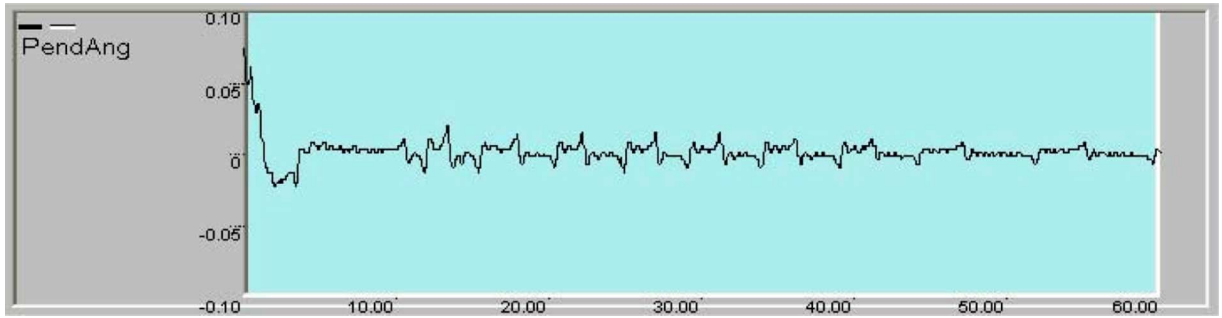


Fig. 14. Experimental result of the pendulum angle.

sampling period $T_s = 0.2$ s and the initial condition $u(t_0) = -4.4$, the control input of system (29) and (30) is given in Fig. 3.

The state vector of the closed-loop system (29) and (30) with initial conditions $x(t_0) = [0 \ 0.2 \ 0 \ 0]^T$ is given in Fig. 4.

It is shown from Fig. 4 that the closed-loop system (29) and (30) with sampling period $T_s = 0.2$ s is divergent. Then, we choose a smaller sampling period $T_s = 0.02$ s. The control input of the closed-loop system (29) and (30) with an initial condition $u(t_0) = -4.4$ is given in Fig. 5.

The state vector of the closed-loop system (29) and (30) with initial conditions $x(t_0) = [0 \ 0.2 \ 0 \ 0]^T$ is given in Fig. 6.

It is shown from Fig. 6 that the closed-loop system (29) and (30) with sampling period $T_s = 0.02$ s is convergent.

From above simulation results, we have that the sampling period $T_s = 0.02$ s is fit for the inverted-pendulum system. Hence, we should use the delta operator approach for sampling system (29) and (30). The following delta operator system is obtained

where

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0.2943 & 1.0020 \\ 0 & 0 & 29.4577 & 0.2943 \end{bmatrix}, \quad B = \begin{bmatrix} -0.0100 \\ -1.0000 \\ 0.0300 \\ 3.0059 \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

In the following, we give the following simulation results to show the effectiveness of the designed delta operator Kalman filter and its convergence analysis. Consider the initial condition of error covariance matrix $P(t_k)$ as follows:

$$P(t_0) = \begin{bmatrix} 1.5 & 2 & 1 & 2 \\ 2 & 1 & 0 & 3 \\ 1 & 0 & 2 & 1 \\ 2 & 3 & 1 & 1 \end{bmatrix}. \quad (35)$$

The estimated state $\hat{x}(t_k)$ in every sampling instant t_k can be gotten for system (33) and (34), with the initial condition $\hat{x}(0) = [0 \ -0.2 \ 0 \ 0]^T$. The estimated state vector is given

$$\delta x(t_k) = Ax(t_k) + Bu(t_k) + \omega(t_k) \quad (33)$$

$$y(t_k) = Cx(t_k) + v(t_k) \quad (34)$$

in Fig. 7, in which $\hat{x}(t_k)$ consists of $xe_1(t_k)$, $xe_2(t_k)$, $xe_3(t_k)$ and $xe_4(t_k)$.

The simulation results of the biggest singular value of error covariance matrix $P(t_k)$ denoted by $\|P(t_k)\|$ is given in Fig. 8.

Fig. 9 shows the convergence rate of the state error norm $\|x(t_k) - \hat{x}(t_k)\|$.

Fig. 10 shows the simulation results of the control input for initial condition $u(0) = 4.4$.

For initial condition: $y(0) = [0\ 0]^T$, the simulation results of the output for system (33) and (34) are given in Fig. 11, where $y(t_k)$ is denoted by $y_1(t_k)$ and $y_2(t_k)$.

It is clearly shown from the above simulation results that the estimated state vector is convergent to the real state vector in this inverted pendulum model. Hence, the delta operator Kalman filter obtained in Theorem 1 is convergent.

In the end, the real-time experiment results are given on the laboratory-scale setup in this section. The control signal of real inverted-pendulum system based on the real-time experiment results is shown in Fig. 12. The cart position of the real inverted-pendulum system is shown in Fig. 13. The pendulum angle of the real inverted-pendulum system is shown in Fig. 14. The model used for control is very simple and neglects effects such as static friction. However, the inverted pendulum control problem is challenging because it may go unstable for sampling frequencies that are not sufficiently high. Therefore, the practical inverted pendulum model is presented as an interesting application of the delta operator Kalman filter in this paper. And the experimental setup in this section has shown the developed techniques of the designed delta operator Kalman filter is practicable.

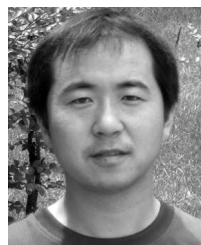
V. CONCLUSION

In this paper, we have considered a delta operator Kalman filter and applied it for an inverted pendulum model in experimental results. The designed delta operator Kalman filter can express both the continuous-time and discrete-time cases. We have also given a new sufficient condition of convergence analysis for the designed delta operator Kalman filter based on Lyapunov functional. On the inverted pendulum experimental results, we have used a delta operator system to describe the practical model and the designed delta operator Kalman filter to estimate the state vectors. The experimental results have shown the effectiveness of the developed techniques. In summary, this paper has given fundamental results for the analysis and application of the designed delta operator Kalman filter.

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